

REFLECTIVE JOURNAL: REGRESSION AND CLASSIFICATION

Course: ITAI1371

Lab Assignment: Lab 06

Group ID: ML_1371_12321_9

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1. Introduction and Objectives

The fundamental objective of this laboratory assignment was to explore, construct, and evaluate supervised machine learning frameworks, focusing explicitly on the core mechanics separating Regression and Classification tasks. Utilizing the historical Titanic passenger dataset, the practical goal was to deploy a LinearRegression model to estimate a continuous numerical variable (Fare) and a LogisticRegression model to predict a binary, categorical class outcome (Survived). As the single, independent operator of group **ML_1371_12321_9**, I managed all steps of this implementation lifecycle autonomously.

2. Breakdown of Individual Contributions

Operating as a solo team member required me to oversee every conceptual, structural, and programming layer of this project. My personal project milestones included:

- **Data Cleaning & Engineering:** I managed the initial pipeline by locating and filtering incomplete records within the Age feature vector. I manually engineered the categorical string feature Sex into a machine-readable binary schema (male: 0, female: 1) to enable optimization within the classification matrices.
- **Model Pipeline Engineering:** I successfully isolated structural matrices (X) and target vectors (y), applied strict structural row constraints, and executed an 80/20 train-test split across both mathematical models to insulate evaluation metrics from data leakage.

- **Troubleshooting & Code Optimization:** I diagnosed and resolved syntax compression issues and script-level environmental constraints within the workspace (specifically, runtime reference errors arising from uninstantiated mathematical libraries like numpy).
 - **Experimental Data Visualization:** I independently conceptualized and appended three custom exploratory data visualization charts utilizing matplotlib and seaborn to audit model prediction patterns, error tracking distributions, and confusion boundaries.
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3. Key Machine Learning Insights & Core Concepts

(Regression) vs. (Classification)

This lab crystalized the structural divide regarding the nature of predicted outputs. Through hands-on application, I proved that regression structures are bound to continuous numerical spectrums answering quantitative metrics ("How much?"), whereas classification structures map variables to rigid boundaries answering qualitative inquiries ("Which category?").

Evaluation Metric Philosophy

A major takeaway was understanding the mathematical incompatibility of cross-applying evaluation metrics. I analyzed why `accuracy_score` is completely useless for a continuous target like Fare. Because accuracy requires a perfect decimal match, a fractional deviation (e.g., predicting \$48.01 for a true \$48.00 value) is calculated as a complete system failure. Conversely, `mean_squared_error` (MSE) provides a realistic continuous performance window by calculating the numerical distance between the regression hyperplane and the ground truth.

Visualizing Model Error Behaviors

By implementing three supplementary validation experiments, I gained valuable insights into model tendencies:

1. **Actual vs. Predicted Plot:** This showed that while the LinearRegression model captures the base pricing trends of higher-class cabins, extreme outliers and high-fare anomalies skew the global fitting line.
2. **Residual Error Distribution:** Plotting the errors (Actual - Predicted) revealed a right-skewed normal curve centered near zero, proving that while the model exhibits stable base assumptions, it consistently underestimates premium-tier accommodations.
3. **Confusion Matrix Analysis:** This matrix provided a clear look at the classification model's trade-offs, pinpointing exactly where passenger demographics failed to align with statistical survival rules (False Positives and False Negatives).

4. Challenges Faced and Technical Resolution

The primary hurdle during this laboratory iteration involved format compression errors. During raw script transitions, code rows collapsed together, triggering critical Python SyntaxError: invalid syntax exceptions because individual variable initializations and independent library packages occupied the same processing lines.

Furthermore, a missing structural dependency for numpy initially blocked the root-mean-square calculation, causing a NameError: name 'np' is not defined. I systematically resolved these issues

by manually restructuring the notebook's grid layers, isolating commands to single rows, and embedding local library packages directly into the active processing scopes.

5. Personal Takeaways and Conclusion

Completing this lab completely solo was highly rewarding. Managing the entire machine learning pipeline independently forced me to understand the underlying theory rather than just copying standard definitions. Moving forward, I plan to explore how data scaling techniques (such as normalization and standardization) can help minimize the residual errors observed in our regression model, and how advanced ensemble models can improve classification accuracy on complex datasets.
